
Zero-Shot Image Retrieval Using Vision-Language Models

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Abstract

This project investigates zero-shot image retrieval using Vision-Language Models (VLMs) by combining text-to-image generation with embedding-based similarity search. The objective is to retrieve relevant images from an unlabeled dataset based on textual queries without relying on annotated training data. A pipeline was implemented where text prompts generate reference images using the Stable Diffusion model, and embeddings are extracted using both CLIP and DINO. These embeddings are compared to those of dataset images using cosine similarity to identify the most semantically similar matches. Evaluation was conducted manually through visual inspection of retrieved results. The findings demonstrate the potential of generative approaches for retrieval tasks in low-label or zero-label environments and highlight differences between embedding methods and their impact on retrieval quality. The code of this project can be found in this Github repository.

1 Introduction and Related Works

Recent advancements in Vision-Language Models (VLMs) have enabled significant progress in tasks that bridge textual and visual domains, such as text-to-image generation and zero-shot image retrieval. Traditional image retrieval systems typically rely on large annotated datasets with explicit labels. However, such annotations are often unavailable or infeasible to obtain in real-world applications. This project addresses the problem of image retrieval in a zero-shot setting, where the goal is to retrieve relevant images from an unlabeled dataset using only a textual query.

Rather than attempting to match a text query directly to existing images, this approach first generates a synthetic image that visually represents the query. This reference image is then embedded using pre-trained models and compared to the embeddings of dataset images to identify the most similar instances. This method provides an alternative path to text-image matching, especially when working with unlabeled or sparse datasets. The workflow of this project can be seen in Figure 1.

1.1 Stable Diffusion

Stable Diffusion[5] is a latent diffusion model that generates high-quality images from textual prompts by operating in a compressed latent space rather than pixel space. This design enables faster generation with reduced memory usage while maintaining high fidelity in outputs. The model is trained on large-scale text-image datasets and can produce detailed and coherent images that align well with complex prompts. Its open-source availability and efficiency make it a practical choice for zero-shot retrieval pipelines requiring dynamic generation.

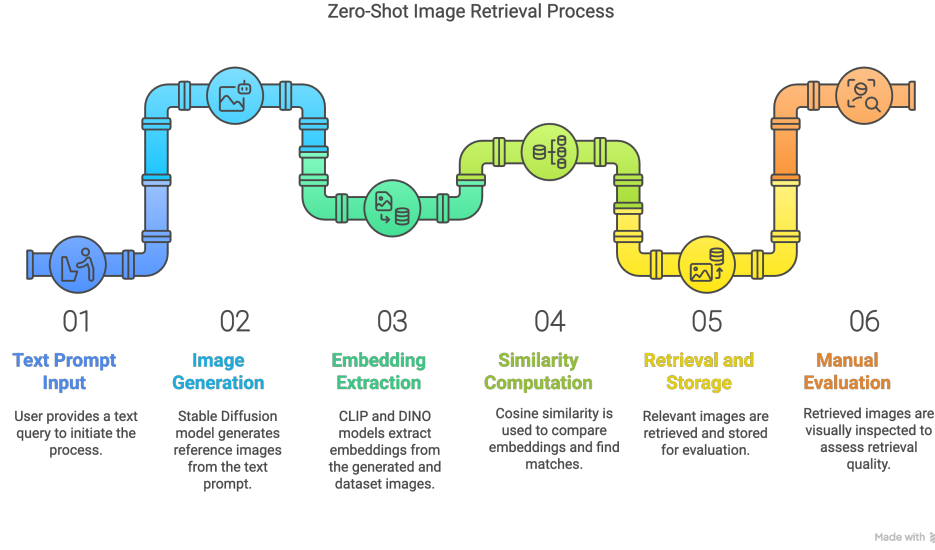


Figure 1: Pipeline for Zero-Shot Image Retrieval. Text prompts are used to generate reference images with Stable Diffusion (v1.5 & v2.1). Embeddings from both generated and dataset images are extracted using CLIP and DINO, followed by cosine similarity comparison to retrieve the most semantically similar matches. Final results are evaluated manually.

1.2 Contrastive Language-Image Pretraining (CLIP)

CLIP[3] is a widely used model that jointly trains image and text encoders to map inputs into a shared semantic space. It was trained on 400 million image-text pairs using a contrastive loss, allowing the model to associate text descriptions with corresponding visual representations. CLIP supports zero-shot learning by enabling similarity comparison between text and image embeddings, which has been applied in classification, retrieval, and captioning tasks without fine-tuning.

1.3 Self-Distillation with No Labels (DINO)

DINO[1] is a self-supervised learning method that trains vision transformers without requiring labeled data. DINO captures semantic features through knowledge distillation, where a student model learns from the outputs of a slowly updated teacher model. The resulting embeddings are not aligned with language, but they are highly effective for visual similarity and clustering tasks. In this project, DINO embeddings offer an alternative perspective to CLIP by focusing purely on visual semantics without language alignment.

2 Methodology

The system is designed to perform image retrieval based on textual queries in a zero-shot setting, without the need for labeled training data. A small subset of real-world images was used as the retrieval target set, consisting of 30 images selected from the COCO[2] dataset. The core idea is to generate a synthetic image from a given prompt and use that image to search for visually similar items within an unlabeled dataset. The methodology consists of four key stages: prompt-driven image generation, embedding extraction, similarity computation, and retrieval.

2.1 Text-to-Image Generation

For each text query, reference images were generated using both Stable Diffusion v1.5 and Stable Diffusion v2.1. These models are latent diffusion models that operate in a compressed latent space, allowing efficient generation of high-resolution images from textual descriptions. Using both versions

allows comparison between generation quality and downstream retrieval performance. The prompts were written manually, with the goal of producing reference images that reflect the semantics of the target dataset.

A visual comparison between COCO dataset image and its corresponding generations using both Stable Diffusion versions is shown in Figure 2. The prompt used for generation was: “*Colorful dessert with banana, chocolate, and marshmallows.*”



Figure 2: Visual comparison between a dataset image and its corresponding generations using Stable Diffusion v1.5 and v2.1. The prompt used for generation was: “*Colorful dessert with banana, chocolate, and marshmallows.*”

2.2 Embedding Extraction

Once the images were generated, visual embeddings were extracted using two different pre-trained models: CLIP ViT-B/32 and DINOv2. CLIP is used to generate embeddings aligned with language semantics, while DINOv2 provides a language-free representation that captures high-level visual structure. Embeddings for the generated images and the dataset images were computed in advance to enable efficient similarity comparisons.

2.3 Similarity Computation

Cosine similarity was used to compare the embedding of each generated image with those of the dataset images. Comparisons were conducted independently for both CLIP-based and DINO-based embeddings, and separately for images generated by Stable Diffusion v1.5 and v2.1. This setup enables retrieval analysis across both generation quality and embedding model type.

2.4 Retrieval and Result Storage

For each prompt, the two most similar images from the dataset were retrieved using each of the four combinations: (1) SD v1.5 + CLIP, (2) SD v1.5 + DINO, (3) SD v2.1 + CLIP, and (4) SD v2.1 + DINO. Retrieving the top two matches was particularly useful for broader prompts that could correspond to multiple images. In addition to the retrieved images, cosine similarity scores were saved to support visual inspection and model comparison. The final output for each prompt includes the generated images, the top two retrieved dataset images per model pair, and their respective similarity scores.

3 Experiments

3.1 Dataset

The retrieval dataset was constructed using a subset of images from the 2017 validation set of the COCO dataset[2] (Common Objects in Context). COCO is a large-scale dataset widely used in computer vision tasks such as object detection, segmentation, and captioning. It contains over 330,000 images annotated with object labels and bounding boxes, making it a strong candidate for retrieval tasks involving real-world scenes.

For this project, 30 images were selected randomly from the validation set. The validation split was chosen due to its smaller size compared to the training set, which made it more manageable for manual evaluation. One condition was applied during selection: each image had to contain exactly two or three distinct objects. This constraint allowed for writing prompts that were creative and open-ended, while still ensuring that the resulting scenes would not be too complex or cluttered. The goal was to maintain a balance between realism and visual clarity, making the retrieval task both meaningful and tractable.

3.2 Prompt Design

A total of 20 prompts were written based on the dataset images. Prompt creation was informed by the content of the selected dataset to ensure relevance and avoid unrealistic mismatches. The prompts were designed with varying levels of specificity:

- 10 prompts were detailed, often corresponding closely to a single image in the dataset. These prompts typically described concrete object arrangements or visual scenes (e.g., “Woman sitting near Eiffel Tower with pigeons”).
- 10 prompts were general, allowing for multiple valid matches across the dataset. These often describe broader scenes or object categories (e.g., “Moments of motion and action”).

This mix of prompt types was intended to test how well the system could handle both precise and ambiguous queries.

3.3 Retrieval Behavior

For each prompt, two images were generated—one using Stable Diffusion v1.5 and one using v2.1. Embeddings for both generated and dataset images were extracted using CLIP and DINOv2. The cosine similarity was computed to identify the top two most similar dataset images for each of the four model combinations.

Retrieval results varied depending on both the generation model and embedding method. In some cases, DINO-based embeddings favored structural or visual layout similarities, while CLIP prioritized semantic alignment. Stable Diffusion v2.1 occasionally produced visually richer or more complex images compared to v1.5, but this did not always translate to improved retrieval quality.

3.4 Evaluation

Evaluation was conducted manually by visually inspecting the retrieved images alongside the generated ones and the original prompts. This allowed assessment of whether the retrieved images reflected the prompt’s semantics and how consistent the results were across different model pairings. No automated metrics were used, due to the unlabeled nature of the dataset.

The label proportions provide insight into the behavior of different models under varying conditions. As shown in Table 1, CLIP generally performed better than DINO, retrieving a higher proportion of related images and fewer not related ones. Table 2 shows that SD v2.1 slightly outperformed v1.5 in retrieval relevance, though the difference was modest. Interestingly, Table 3 indicates that general prompts led to better retrieval alignment than detailed ones, with higher related proportions and fewer mismatches. This suggests that the system handled broader semantic cues more robustly than very specific descriptions.

Table 1: Label proportions by embedding model

Embedding Model	Related	Somehow related	Not related
CLIP	0.700	0.075	0.225
DINO	0.650	0.100	0.250

To better understand how different combinations of models and prompts affected retrieval outcomes, retrieved images were labeled into three categories: **Related**, **Somehow related**, and **Not related**.

Table 2: Label proportions by SD version

SD Version	Related	Somehow related	Not related
SD v1.5	0.6625	0.0875	0.250
SD v2.1	0.6875	0.0875	0.225

Table 3: Label proportions by prompt type

Prompt Type	Related	Somehow related	Not related
Detailed	0.625	0.100	0.275
General	0.725	0.075	0.200

Figure 3 shows label distributions across different embedding models (CLIP and DINO) and prompt types (detailed vs. general). Figure 4 presents the same distributions across image generation models (Stable Diffusion v1.5 and v2.1).

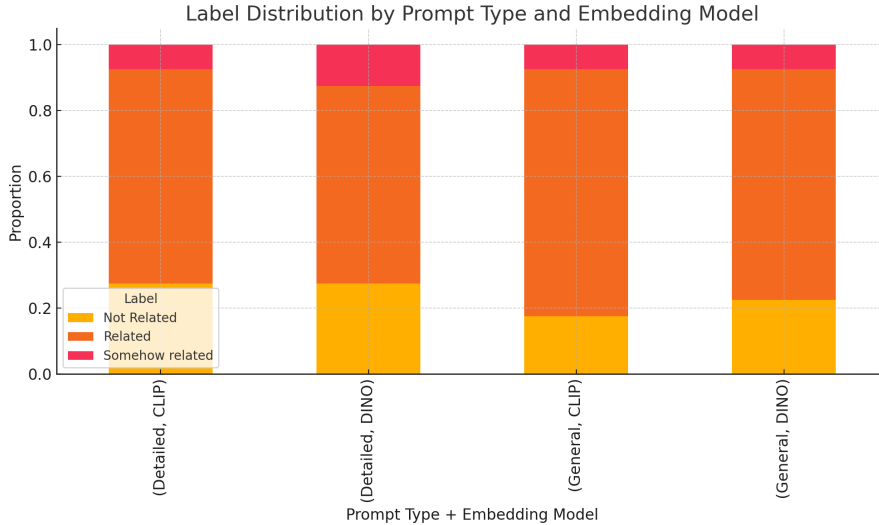


Figure 3: Label distribution of retrieval results based on prompt type and embedding model. CLIP tended to retrieve semantically aligned images, especially for general prompts.

4 Conclusion and Future Work

This project explored a generative approach to zero-shot image retrieval by combining text-to-image synthesis with embedding-based similarity search. Using Stable Diffusion models to generate reference images and CLIP and DINOv2 for visual embedding, the system retrieved the most semantically similar images from an unlabeled subset of the COCO dataset. Manual evaluation demonstrated that the pipeline can produce meaningful matches, especially when the prompts and target images were aligned in complexity and content.

While the approach showed promise, the project was constrained by several practical limitations. The dataset was deliberately kept small (30 images) to enable manual analysis, but this limited the diversity of retrieval outcomes. The number of prompts (20) was also modest, which restricted the scope of evaluation. On the computational side, limited GPU resources imposed constraints on the number of generated images and the ability to test additional diffusion models or larger embedding architectures. As a result, the retrieval behavior was assessed using only a few combinations of generation and embedding models.

Future work could expand on this foundation in several directions. Increasing the dataset size and prompt variety would allow for more comprehensive evaluation, possibly incorporating human-in-the-

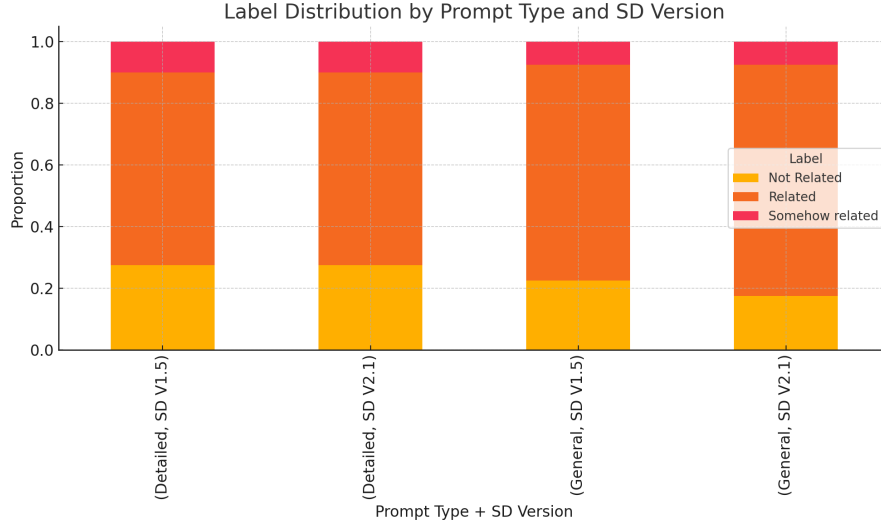


Figure 4: Label distribution of retrieval results based on prompt type and image generation model. Differences between Stable Diffusion v1.5 and v2.1 were relatively minor, with v2.1 slightly favoring broader scene coverage.

loop feedback or crowd-sourced annotations. Additional generation models, such as Kandinsky[4] or Imagen[6], could be explored for prompt diversity. On the embedding side, testing newer multimodal or fine-tuned encoders might yield more robust retrieval behavior. Finally, incorporating lightweight evaluation metrics, even approximate ones, could help quantify performance at a larger scale.

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